History of Java-Environmental Setup -features of java-data types- variables- modifiers-keywords operators- Iterative, Conditional and control statement - command line arguments - string- string buffer - simple java program- enumerators-array-formatting output.

History of Java & Environmental Setup

1. Java was originally developed by _____ at Sun Microsystems.
   Answer: James Gosling
2. Java was originally called _____.
   Answer: Oak
3. Java was officially released by Sun Microsystems in the year _____.
   Answer: 1995
4. The Java compiler converts source code into _____.
   Answer: Bytecode
5. The Java bytecode is executed by the _____.
   Answer: JVM
6. JDK stands for _____.
   Answer: Java Development Kit
7. JRE stands for _____.
   Answer: Java Runtime Environment
8. JVM stands for _____.
   Answer: Java Virtual Machine

Features of Java

9. Java is a _____ programming language because it supports objects and classes.
   Answer: Object-oriented
10. Java is _____ independent because Java bytecode can run on any system having a JVM.
    Answer: Platform
11. Java supports the concept of _____, which allows the same bytecode to run on different platforms.
    Answer: WORA (Write Once, Run Anywhere)
12. Java provides automatic _____ management.
    Answer: Garbage
13. Java is a _____ typed language.
    Answer: Statically
14. Java supports _____ programming, allowing multiple tasks to execute concurrently.
    Answer: Multithreaded

Data Types, Variables & Modifiers

15. The number of primitive data types in Java is _____.
    Answer: 8
16. The data type used to store a single character is _____.
    Answer: char
17. The data type used to store true or false values is _____.
    Answer: boolean
18. The default value of a boolean instance variable is _____.
    Answer: false
19. The data type used to store decimal numbers with single precision is _____.
    Answer: float
20. A variable declared inside a method is called a _____ variable.
    Answer: Local
21. The keyword used to declare a constant variable is _____.
    Answer: final
22. The modifier that allows a member to be accessed from anywhere is _____.
    Answer: public
23. The modifier that restricts access to the same class is _____.
    Answer: private
24. The _____ modifier indicates that a member belongs to the class rather than to objects.
    Answer: static

Keywords & Operators

25. Words that have predefined meanings in Java are called _____.
    Answer: Keywords
26. The keyword used to inherit a class is _____.
    Answer: extends
27. The keyword used to implement an interface is _____.
    Answer: implements
28. The operator == is called the _____ operator.
    Answer: Equality
29. The operator && is called the logical _____ operator.
    Answer: AND
30. The operator % is called the _____ operator.
    Answer: Modulus
31. The ++ operator is known as the _____ operator.
    Answer: Increment

Conditional, Iterative & Control Statements

32. The statement used to execute a block of code based on a condition is _____.
    Answer: if
33. The statement used to select one option from multiple choices is _____.
    Answer: switch
34. The _____ loop executes a block of code as long as a condition is true.
    Answer: while
35. The _____ loop executes its body at least once.
    Answer: do-while
36. The for statement is commonly used for _____.
    Answer: Iteration
37. The break statement is used to _____ a loop or switch statement.
    Answer: Terminate
38. The continue statement is used to skip the current _____ of a loop.
    Answer: Iteration

Command Line Arguments

39. Command-line arguments are received by the _____ method.
    Answer: main()
40. The parameter used to receive command-line arguments is an array of type _____.
    Answer: String
41. The first command-line argument is accessed using _____.
    Answer: args[0]

String & StringBuffer

42. In Java, String objects are _____.
    Answer: Immutable
43. The class used to create a mutable sequence of characters is _____.
    Answer: StringBuffer
44. The method used to add characters to a StringBuffer is _____.
    Answer: append()
45. The method used to reverse the contents of a StringBuffer is _____.
    Answer: reverse()

Simple Java Program & Enumerators

46. The entry point of a Java application is the _____ method.
    Answer: main()
47. The keyword used to define an enumeration is _____.
    Answer: enum
48. An enumeration is used to represent a fixed set of _____.
    Answer: Constants

Arrays & Formatting Output

49. An array in Java is used to store multiple values of the _____ data type.
    Answer: Same
50. The method commonly used for formatted output in Java is _____.
    Answer: printf()

Object-class-constructor-benefits of OOPS-concepts of OOPS- inheritance - polymorphism-abstract class generic class- Overriding-Overloading-Interface: Implementation of interface-extending interface-inner class- static and dynamic binding- package: Package as Access Protection-CLASSPATH setting- Import packages.

Object, Class & Constructor

1. A class is a _____ or template used to create objects.
   Answer: Blueprint
2. An instance of a class is called an _____.
   Answer: Object
3. The keyword used to create an object in Java is _____.
   Answer: new
4. A special method that is automatically invoked when an object is created is called a _____.
   Answer: Constructor
5. A constructor has the same name as the _____.
   Answer: Class
6. A constructor does not have a _____ type.
   Answer: Return
7. A constructor with no parameters is called a _____ constructor.
   Answer: Default
8. A constructor that accepts parameters is called a _____ constructor.
   Answer: Parameterized

Benefits & Concepts of OOP

9. OOP stands for _____.
   Answer: Object-Oriented Programming
10. The four major concepts of OOP are encapsulation, inheritance, polymorphism and _____.
    Answer: Abstraction
11. Wrapping data and methods into a single unit is called _____.
    Answer: Encapsulation
12. Hiding implementation details and showing only essential features is called _____.
    Answer: Abstraction
13. The ability of an object to take many forms is called _____.
    Answer: Polymorphism

14. The mechanism by which one class acquires the properties of another class is called _____.
    Answer: Inheritance
15. OOP promotes code _____ and reusability.
    Answer: Modularity

Inheritance

16. The keyword used to inherit a class in Java is _____.
    Answer: extends
17. A class that is inherited from is called the _____ class.
    Answer: Superclass
18. A class that inherits another class is called a _____ class.
    Answer: Subclass
19. Java supports _____ inheritance through classes.
    Answer: Single
20. Java does not support multiple inheritance through _____.
    Answer: Classes
21. Multiple inheritance can be achieved in Java using _____.
    Answer: Interfaces

Polymorphism

22. Polymorphism means _____ forms.
    Answer: Many
23. Method _____ is an example of compile-time polymorphism.
    Answer: Overloading
24. Method _____ is an example of runtime polymorphism.
    Answer: Overriding
25. In method overriding, the method in the subclass must have the same _____ as the superclass method.
    Answer: Signature

Abstract & Generic Classes

26. A class declared using the keyword _____ is called an abstract class.
    Answer: abstract
27. An abstract class cannot be directly _____.
    Answer: Instantiated
28. A method without a body is called an _____ method.
    Answer: Abstract
29. A class that can work with different data types using type parameters is called a _____ class.
    Answer: Generic
30. The symbol commonly used to specify a generic type parameter is _____.
    Answer: <T>

Overriding & Overloading

31. Defining a method in a subclass with the same signature as the superclass method is called method _____.
    Answer: Overriding
32. Defining multiple methods with the same name but different parameter lists is called method _____.
    Answer: Overloading
33. Method overloading is resolved at _____ time.
    Answer: Compile
34. Method overriding is generally resolved at _____ time.
    Answer: Runtime
35. The @Override annotation is used to indicate that a method is being _____.
    Answer: Overridden

Interface

36. An interface is declared using the keyword _____.
    Answer: interface
37. A class uses the keyword _____ to implement an interface.
    Answer: implements
38. An interface can be _____ by another interface.
    Answer: Extended
39. A class can implement _____ interfaces.
    Answer: Multiple
40. Interface methods that are abstract by default are implicitly _____ and public in traditional Java interfaces.
    Answer: Abstract
41. A class that implements an interface must provide implementations for its _____ methods, unless the class is abstract.
    Answer: Abstract

Inner Class

42. A class defined inside another class is called an _____ class.
    Answer: Inner
43. An inner class can access members of its _____ class.
    Answer: Outer
44. An inner class that is declared using the static keyword is called a _____ nested class.
    Answer: Static

Static & Dynamic Binding

45. Binding that occurs during compile time is called _____ binding.
    Answer: Static

46. Binding that occurs during runtime is called _____ binding.
    Answer: Dynamic
47. Method overloading generally uses _____ binding.
    Answer: Static
48. Method overriding generally uses _____ binding.
    Answer: Dynamic

Package, CLASSPATH & Import

49. A package in Java is used to group related _____ and interfaces.
    Answer: Classes
50. The keyword used to access classes from another package is _____.
    Answer: import

Exceptions-benefits of exception-Types of Exceptions-Errors-Control flow- JVM reaction to Exception usage of try, catch, throw, final and finally keyword-rethrowing exceptions, exception specification, built in exceptions-File I/O: Standard Streams-Reading and writing Streams- Byte Array Stream-Data Stream- File Stream- Input and output Stream

Exceptions

1. An exception is an abnormal condition that disrupts the normal _____ of a program.
   Answer: Flow
2. Exception handling is used to handle _____ conditions during program execution.
   Answer: Runtime
3. One major benefit of exception handling is that it prevents abnormal program _____.
   Answer: Termination
4. Exceptions in Java are objects derived from the class _____.
   Answer: Throwable
5. The two main subclasses of Throwable are _____ and Exception.
   Answer: Error
6. Exceptions that are checked by the compiler are called _____ exceptions.
   Answer: Checked
7. Exceptions that occur during runtime and are not checked by the compiler are called _____ exceptions.
   Answer: Unchecked
8. Runtime exceptions are subclasses of the _____ class.
   Answer: RuntimeException
9. Errors generally indicate serious problems that applications usually cannot _____.
   Answer: Handle

Control Flow & JVM Reaction

10. When an exception occurs, the JVM creates an exception _____.
    Answer: Object
11. The JVM searches for a suitable _____ block to handle the exception.
    Answer: catch
12. If an exception is not handled, it propagates up the _____ stack.
    Answer: Call
13. The process of transferring control from the point where an exception occurs to an exception handler is called exception _____.
    Answer: Propagation
14. If no appropriate handler is found, the JVM terminates the current _____.
    Answer: Thread

try, catch, throw, finally & final

15. The code that may generate an exception is placed inside the _____ block.
    Answer: try
16. The block used to handle an exception is called the _____ block.
    Answer: catch
17. A try block must be followed by at least one catch or _____ block.
    Answer: finally
18. The keyword used to explicitly generate an exception is _____.
    Answer: throw
19. The keyword used to declare exceptions that a method may pass to its caller is _____.
    Answer: throws
20. The _____ block is normally executed whether an exception occurs or not.
    Answer: finally
21. The final keyword can be used with variables, methods and _____.
    Answer: Classes
22. A final variable cannot be _____ after initialization.
    Answer: Reassigned
23. A final method cannot be _____ by a subclass.
    Answer: Overridden
24. A final class cannot be _____.
    Answer: Inherited

Rethrowing & Exception Specification

25. Throwing an already caught exception again is called _____ an exception.
    Answer: Rethrowing
26. The throws clause provides the exception _____ of a method.
    Answer: Specification

27. A method can declare multiple exceptions in its throws clause separated by _____.
    Answer: Commas

Built-in Exceptions

28. Division by zero for integer values causes _____.
    Answer: ArithmeticException
29. Accessing an invalid array index causes _____.
    Answer: ArrayIndexOutOfBoundsException
30. Calling a method using a null reference can cause _____.
    Answer: NullPointerException
31. Converting an invalid string to a numeric type may cause _____.
    Answer: NumberFormatException
32. Attempting to access an invalid position in a collection can cause an _____ exception.
    Answer: IndexOutOfBoundsException

File I/O & Standard Streams

33. I/O stands for _____.
    Answer: Input/Output
34. The standard input stream in Java is _____.
    Answer: System.in
35. The standard output stream in Java is _____.
    Answer: System.out
36. The standard error stream in Java is _____.
    Answer: System.err
37. The class used for byte-oriented input streams is _____.
    Answer: InputStream
38. The class used for byte-oriented output streams is _____.
    Answer: OutputStream

Reading & Writing Streams

39. The read() method is commonly used to _____ data from an input stream.
    Answer: Read
40. The write() method is commonly used to _____ data to an output stream.
    Answer: Write
41. A stream that transfers data one byte at a time is called a _____ stream.
    Answer: Byte
42. InputStream is an _____ class.
    Answer: Abstract

Byte Array, Data & File Streams

43. The class used to read bytes from a byte array is _____.
    Answer: ByteArrayInputStream

44. The class used to write bytes into a byte array is _____.
    Answer: ByteArrayOutputStream
45. The classes used for reading and writing primitive data types are _____ and DataOutputStream.
    Answer: DataInputStream
46. DataInputStream provides methods such as readInt() and _____.
    Answer: readDouble()
47. The class used to read data from a file as bytes is _____.
    Answer: FileInputStream
48. The class used to write data to a file as bytes is _____.
    Answer: FileOutputStream
49. FileInputStream and FileOutputStream are examples of _____ streams.
    Answer: File
50. The base class for character input streams is _____.
    Answer: Reader